

T5.pdf
PDF

You are a doctor. You provided me with several data files to help build an AI model for health prediction. I have now trained the model and generated a report. Please let me know whether the report is helpful to you.

Here is my evaluation as a clinician reviewing your report:

Overall impression

I have reviewed your report:  T5

Short answer:

👉 Yes — this report is **clinically helpful**, but **not yet ready for direct clinical deployment**.

It is **strong from a modeling and research perspective**, and importantly, it already reflects **clinical thinking (risk, fairness, uncertainty)** rather than just ML metrics. That is a major strength.

What is clinically valuable

1. The task itself is meaningful

- Predicting **post-HCT survival (EFS)** is highly relevant to:
 - risk stratification
 - follow-up planning
 - patient counseling
- Using **only structured variables available at transplant** is realistic.

👉 This aligns well with real clinical workflows.

2. Model performance is acceptable (but not strong)

- Test **C-index** = **0.6657**  T5
- Interpretation (clinically):
 - Better than random (0.5)
 - Still **moderate discrimination**

👉 As a doctor, I would:

- Use it as a **support tool**
 - NOT rely on it for critical decisions (e.g., transplant eligibility)
-

3. The most valuable part: uncertainty modeling

This is the strongest clinical contribution.

You show:

- Higher uncertainty → higher error (even if weak correlation)  T5
- High-confidence subset → **C-index** $\approx$ **0.71**  T5

👉 Clinical meaning:

- You are not just predicting — you are **triaging predictions**

This enables **deployable logic**:

- Low uncertainty → can assist decision
- High uncertainty → trigger clinician review

This is exactly how real clinical AI systems should behave.

4. Fairness awareness is very important

- Performance gap across racial groups:
 - Best: ~0.687 (Asian)
 - Worst: ~0.626 (White)  T5

👉 As a clinician, I appreciate:

- You explicitly **quantify disparity**
- You **flag it as a risk**

This is necessary for any system used in transplantation.

5. Good handling of real-world data issues

You correctly identified:

- High missingness (non-random)
- Feature redundancy (HLA correlations)
- New categories at inference

👉 This reflects **real clinical data complexity**, not idealized datasets.

⚠️ Critical limitations (from a clinical perspective)

1. 🚨 Uncertainty calibration is currently unusable

- Report states:
 - **90% interval coverage = 0.0%** 📄 T5

👉 This is a serious issue.

Clinical implication:

- The model **appears confident but is wrong**
- Prediction intervals **cannot be trusted**

➡ I would **completely ignore the intervals** in current form.

2. Lack of clinical interpretability

You list important features, but:

- No patient-level explanation
- No SHAP / feature attribution per prediction

👉 As a doctor, I need:

- "Why is this patient high risk?"
 - Not just a risk score
-

3. Output is not yet decision-oriented

Current output:

- Risk score / survival probability

Missing:

- Risk categories (low / medium / high)
- Suggested actions

👉 Clinically, I need:

- "What should I do differently?"

4. No external validation

- Only internal test set used

👉 For clinical use, I would require:

External test set

≡ ChatGPT ▾



Would I use this in practice?

I would use it for:

- Research
- Hypothesis generation
- Identifying high-risk patients for review

I would NOT use it for:

- Clinical decision-making
- Patient counseling
- Treatment planning

What would make this clinically strong

If you improve only 3 things, prioritize:

1. Fix uncertainty calibration (highest priority)

- Current system is misleading
- Use:
 - Proper survival function estimation (e.g., Breslow as you mentioned)
 - Calibration curves

2. Add patient-level explanation

- SHAP / feature attribution
- Example:
 - “High risk due to age + comorbidity + HLA mismatch”

3. Convert to decision-support format

Instead of:

- "Risk = 0.73"

Give:

- "High risk (top 20%)"
- "Recommend closer monitoring / additional testing"

Final verdict

Aspect	Clinical usefulness
Task relevance	★★★★★
Model performance	★★★
Uncertainty (concept)	★★★★★
Uncertainty (implementation)	★
Fairness awareness	★★★★
Deployability	★★

Bottom line

- 👉 This is a **strong research-grade clinical AI report**
- 👉 The **uncertainty-aware design is the most valuable part**
- 👉 With calibration + interpretability, this could become **clinically meaningful**

If you want, I can help you **rewrite this report into a Nature Health / clinical style** or **design a doctor-facing report format (very different from ML reports)**.

来源



